

1. For each matrix equality, give the elementary coefficient matrix that enables to go from the right to the left. You can either multiply by the elementary matrix on the left or on the right but not on both sides.

(a) ($L_1 \leftarrow L_1/2$)

$$\begin{pmatrix} 2 & -1 & -1 & -4 \\ 3 & 5 & -2 & -4 \end{pmatrix} = \begin{pmatrix} 4 & -2 & -2 & -8 \\ 3 & 5 & -2 & -4 \end{pmatrix}$$

(b) ($L_1 \leftarrow L_1 + L_2$)

$$\begin{pmatrix} 0 & 1 & -15 & 0 & 3 \\ 8 & 9 & -8 & -9 & 5 \\ 5 & -1 & 10 & -2 & -6 \end{pmatrix} = \begin{pmatrix} -8 & -8 & -7 & 9 & -2 \\ 8 & 9 & -8 & -9 & 5 \\ 5 & -1 & 10 & -2 & -6 \end{pmatrix}$$

(c) ($L_2 \leftrightarrow L_4$)

$$\begin{pmatrix} -6 & -8 & 1 & -10 \\ 10 & -7 & 7 & -2 \\ -7 & -2 & -6 & -2 \\ -5 & 5 & 3 & -8 \\ -9 & -7 & 9 & 1 \end{pmatrix} = \begin{pmatrix} -6 & -8 & 1 & -10 \\ -5 & 5 & 3 & -8 \\ -7 & -2 & -6 & -2 \\ 10 & -7 & 7 & -2 \\ -9 & -7 & 9 & 1 \end{pmatrix}$$

(d) ($C_1 \leftarrow C_1 + C_4$)

$$\begin{pmatrix} 1 & -10 & 0 & 5 & -5 \\ 7 & -5 & 7 & 4 & 5 \\ 0 & 3 & 4 & -1 & -3 \end{pmatrix} = \begin{pmatrix} -4 & -10 & 0 & 5 & -5 \\ 3 & -5 & 7 & 4 & 5 \\ 1 & 3 & 4 & -1 & -3 \end{pmatrix}$$

(e) ($C_1 \leftrightarrow C_3$)

$$\begin{pmatrix} -5 & -5 & 6 \\ -1 & -6 & 3 \\ 6 & -10 & 8 \\ 5 & 0 & 9 \\ 2 & -1 & -1 \end{pmatrix} = \begin{pmatrix} 6 & -5 & -5 \\ 3 & -6 & -1 \\ 8 & -10 & 6 \\ 9 & 0 & 5 \\ -1 & -1 & 2 \end{pmatrix}$$

2. Give the inverse of the following matrices. If you think the matrix is not invertible, that might be possible.

(a)

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}^{-1} = \begin{pmatrix} & & & \\ & & & \\ & & & \\ & & & \end{pmatrix}$$

(b)

$$\begin{pmatrix} 1 & 0 & 2 & 5 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}^{-1} = \begin{pmatrix} & & & \\ & & & \\ & & & \\ & & & \end{pmatrix}$$

(c)

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \\ 0 & 0 \end{pmatrix}^{-1} = \begin{pmatrix} & \\ & \\ & \\ & \end{pmatrix}$$

(d)

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}^{-1} = \begin{pmatrix} & & & \\ & & & \\ & & & \\ & & & \end{pmatrix}$$

(e)

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}^{-1} = \begin{pmatrix} & & & \\ & & & \\ & & & \\ & & & \end{pmatrix}$$

(f)

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & \frac{1}{2} \end{pmatrix}^{-1} = \begin{pmatrix} & & & \\ & & & \\ & & & \\ & & & \end{pmatrix}$$

3. Number of operations for this algorithm. (Only first order term, some explanations needed.)

```
def upper_triangular_times_square(A,B):  
    n = A.shape[0]  
    C = np.zeros((n,n))  
    for i in range(0,n):  
        for j in range(0,n):  
            for k in range(i,n):  
                C[i,j] = C[i,j] + A[i,k] * B[k,j]  
  
    return C
```

Note:

$$\sum_{k=1}^n k = \frac{n(n+1)}{2}$$

$$\sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6}$$

$$\sum_{k=1}^n k^3 = \left(\frac{n(n+1)}{2} \right)^2$$